

MBUA 512: Data, Analytics & Insights

Class 3: SQL (Michael Winikoff)

(SQL = Structured Query Language, "Es-Queue-El" or "Sequel")

Recap from last week (database design):

1. Data modelling using diagrams
2. Translating diagrams to relational databases and the role of *keys*
3. Anomalies and avoiding them by *normalisation*

These slides are based on slides by Professor Markus Luczak-Roesch 

Some useful superset commands

```
show schemas from idrive  
show tables in idrive.relbench_rel_ratebeer – most tables  
show tables in idrive.relbench_rel_avito  
select * from idrive.relbench_rel_ratebeer.countries
```

Select Schema `user_[username]` replacing `[username]` with your username (e.g. for me `user_michaelwinikoff`)

SQL commands

- `CREATE TABLE` and `ALTER TABLE` - define the structure of a table (its *schema*)
- `INSERT INTO` and `DELETE FROM` and `UPDATE` - change the data in the table
- `SELECT` - query data

(there are a lot more than these)

CREATE TABLE

```
CREATE TABLE name {  
    column_name data_type,  
    column_name data_type  
    ...  
}
```

QUESTION: HOW DEAL WITH KEYS?!

QUESTION: in order to keep DB creation in the first small assignment need to cover in week 1 and/or 2

Trino types

- Boolean (`true` or `false`)
- Integer or Int (`12` , `-14`)
- Real, Double, Number (`3.14` , `-9.8345`) [floating point, inexact]
- Decimal (3.29) [exact up to 38 digits]
- Varchar - a string (`'Hello!'`) ["var" = variable length, "char" = character, note: use single quote!]
- Varbinary - binary data
- Date (`DATE '2026-12-31'`) [note: year-month-day with the word "DATE"]
- Time (`TIME '01:58:36.77'`) (note the word "TIME")
 - also have a time with time zone

NOTE: for MySQL don't use "DATE" word ... and need a length for

<https://trino.io/docs/current/sql/create-table.html>

```
CREATE TABLE countries (  
    code VARCHAR(50),  
    name VARCHAR(50)  
);
```

ALTER TABLE

ALTER TABLE - change the definition of an existing table

e.g. ALTER TABLE name ADD COLUMN column_name data_type

ALTER TABLE countries ADD COLUMN continent Varchar

<https://trino.io/docs/current/sql/alter-table.html>

INSERT INTO

INSERT INTO table_name VALUES (column, column, column) (val,val,val) ...

IMPORTANT: single quote are for strings, double for DB objects.

```
INSERT INTO countries (code, name, continent) VALUES  
( 'A7', 'Kepler Dome', 'Moon'),  
( 'A8', 'Blip', 'Moon'),  
( 'A9', 'Blah', 'Moon'),  
( 'A10', 'Blip', 'Moon'),  
( 'A11', 'Blah', 'Moon'),  
( 'A12', 'Blip', 'Moon');
```

Note: how to generate from CSV - ask LLM (e.g. CoPilot, Claude, local Gemma) or use Spreadsheet with a few formulae (next slide)

Using a spreadsheet ...

code	name	continent				INSERT INTO countries VALUES
A7	Kepler Dome	Moon	'A7'	'Kepler Dome'	'Moon'	('A7','Kepler Dome','Moon'),
A8	Blip	Moon	'A8'	'Blip'	'Moon'	('A8','Blip','Moon'),
A9	Blah	Moon	'A9'	'Blah'	'Moon'	('A9','Blah','Moon'),
A10	Blip	Moon	'A10'	'Blip'	'Moon'	('A10','Blip','Moon'),
A11	Blah	Moon	'A11'	'Blah'	'Moon'	('A11','Blah','Moon'),
			=""&A2&""			="(" & TEXTJOIN(", ",FALSE,D2:F2, & "),"

DELETE FROM table

SELECT

```
SELECT what  
FROM table(s)  
WHERE conditions
```

```
-- Example - note "--" starts a comment to the end of the line  
/* This is also  
a comment  
*/  
SELECT * -- "*" means "everything"  
FROM relbench_rel_ratebeer.countries  
WHERE has_brewer = false
```

<https://trino.io/docs/current/sql/select.html>

By far the most complex - will cover basics today, and more next week

Today: SQL

- `CREATE TABLE` and `ALTER TABLE` - define the structure of a table (its *schema*)
- `INSERT INTO` and `DELETE FROM` and `UPDATE` - change the data in the table
- `SELECT` - query data

Next:

- Activities in part 2 today - learn by doing!
- Next week: more SQL - focusing on `SELECT`

Activities

 **Activity: ... (~10 minutes, in groups)**